INCH-POUND
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(See 6.8)

MILITARY SPECIFICATION

LAUNDRY PRESSING AND FINISHING EQUIPMENT MODIFIED COMMERCIAL (SHIPBOARD)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers laundry pressing and finishing equipment for surface shipboard use.

1.2 Classification. Laundry pressing and finishing equipment shall be of the following types and classes, as specified (see 6.2.1):

Type I – Laundry presses, steam heated
- Class B - Trouser topping, double
- Class C - General utility
- Class G - Collar-yoke-cuff press (two head)
- Class N - Double sleeve, noncabinet, nonbag type

Type II – Laundry presses, electrically heated
- Class B-E - Trouser topping, double
- Class C-E - General utility
- Class G-E - Collar-yoke-cuff press (two head)
- Class N-E - Double sleeve, noncabinet, nonbag type

Type III – Flatwork ironer, cylinder
- Class A – 85-inch ironer
- Class B – 60-inch ironer.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3510

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-F-320	Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes

MILITARY

MIL-E-917	Electric Power Equipment, Basic Requirements (Naval Shipboard Use)
MIL-L-3153	Laundry and Dry Cleaning Machinery and Equipment, Packaging of
MIL-S-8660	Silicone Compound, NATO Code Number S-736
MIL-M-17060	Motors, 60-Hertz, Alternating Current, Integral-Horsepower. Shipboard Use
MIL-L-17331	Lubricating Oil, Steam Turbine and Gear, Moderate Service
MIL-L-19140	Lumber and Plywood, Fire-Retardant Treated
MIL-G-23549	Grease, General Purpose
DOD-G-24508	Grease, High Performance, Multi-purpose (METRIC)

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-167-1	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of

(Copies of specifications and standards, required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

Y14.5M Dimensioning and Tolerancing

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Section II
Material Specifications, Part A - Ferrous
SA 53 (Grade B) Specification for Welded and Seamless Pipe

Section VII
Boiler and Pressure Vessel Code

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 36	Standard Specification for Structural Steel; (DOD adopted)
A 47	Standard Specification for Ferritic Malleable Iron Castings; (DOD adopted)
A 167	Standard Specification for Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip; (DOD adopted)
A 269	Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service; (DOD adopted)
B 61	Standard Specification for Steam or Valve Bronze Castings; (DOD adopted)
B 62	Standard Specification for Composition Bronze or Ounce Metal Castings; (DOD adopted)
B 75	Standard Specification for Seamless Copper Tube; (DOD adopted)
B 117	Standard Method of Salt Spray (Fog) Testing; (DOD adopted)
B 127	Standard Specification for Nickel-Copper Alloy (UNS NO4400) Plate, Sheet, and Strip; (DOD adopted)
B 209	Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate, (DOD adopted)

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D 3359	Standard Methods for Measuring Adhesion By Tape Test, (DOD adopted)
D 3363	Standard Test Method For Film Hardness By Pencil Test
1166-88	Human Engineering Design Criteria for Marine Systems, Equipment and Facilities; (DOD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

SG-3	Low-Voltage Power Circuit Breakers; (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturer's Association, 2101 L St., NW, Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

National Electric Code Handbook

(Applications for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

AS604	Hose Assembly, Tetrafluoroethylene, 400 Degree F, 3000psi Hydraulic, Heavy Braid (ANSI Approved) (DOD Adopted)
AS620	High Temperature Hose Assembly, Convoluted Tetrafluoroethylene, for Aircraft (ANSI Approved)

(Applications for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Nongovernment standard and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.1.1 Current design of equipment. Laundry pressing and finishing equipment shall be new and furnished complete with motors, drive mechanism, starters and controllers, including master switches, timers, reversing mechanism, and other electrical equipments.

3.12 Human factors engineering. Human factors engineering criteria, principles, and practices, as defined in ASTM 1166, shall be used in the construction of the laundry pressing and finishing equipment as specified herein. The equipment shall be such that access for operation, maintenance, and repair shall be analyzed in terms of the applicable requirements of ASTM 1166. Access to each control, valve, lubrication fitting, or any other component shall include space for any tools or test equipment required, as well as any protective clothing likely to be worn by the maintainer. Maintenance and operational activities shall permit performance by the fifth percentile female and ninety-fifth percentile male in accordance with ASTM 1166. Particular attention shall be given to the ease of access for replacement of bearings, adjustment of belt and chain drives, and lubrication of bearings and chain drives. Controls required for the maintenance and operation of this equipment shall be as specified in 3.3.20. Skill level and training requirements for operation, maintenance. and repair as influenced by the machine design shall be kept at the lowest level consistent with other requirements herein. The analysis of operation, maintenance, and repair functions shall include and reflect considerations of safety.

3.2 Material.

3.2.1 Corrosion-resisting metal. Corrosion-resisting metal shall be corrosion-resisting steel in accordance with ASTM A 167 or ASTM A 269 type 304, or nickel-copper alloy in accordance with ASTM B 127. The use of cadmium in any form is prohibited.

3.2.2 Metal joints. Rivets, bolts, machine screws, nuts, and washers shall be of steel, except when brass or corrosion-resisting metal is fastened, in which case fasteners shall be of brass or corrosion-resisting metal respectively. Where dissimilar metals are fastened, rivets, bolts, machine screws, nuts, and washers shall be galvanically compatible with materials being joined. Electrodes or welding rods used for welding shall deposit material similar to the base metal. Nuts subject to vibration shall be adequately locked. The use of drill point screws shall be prohibited.

3.2.3 Piping, fittings, valves, and connections.

3.2.3.1 Pipes. Pipes shall be made of corrosion-resisting steel, grade TP 304 in accordance with ASTM A 269 or copper in accordance with ASTM B 75.

3.2.3.2 Fittings. Fittings shall be made of bronze in accordance with ASTM B 61 or ASTM B 62.

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3.2.3.3 Valves. Valves shall be made of cast bronze in accordance with ASTM B 61. The trim (stem, disc, and seat) shall be of corrosion-resisting steel. Valves shall function in the temperatures and pressures as specified herein.

3.2.4 Conduit and wire. Electrical conduit shall be either flexible steel, zinc-coated conduit or electrical metallic tubing. Wire shall be stranded annealed copper wire, sized properly for current loading, insulated to withstand press heat flux, and shielded to reduce electromagnetic interference emissions.

3.2.5 Structure. Frame shall be of malleable iron in accordance with ASTM A 47, steel in accordance dance with ASTM A 36, or of a steel of equal or better yield strength and ultimate strength, finished for corrosion resistance. Gray east iron or other brittle materials shall not be used for structural parts. Gray cast iron may be used for components like gear reducers and bearing housings which are not strength members unless manufactured as an integral part of the frame. No aluminum castings shall be permitted.

3.2.6 Head and buck. Heads and bucks shall be made of ASTM A 36 steel, ASTM A 47 malleable iron, or of steel or iron with equal or better yield strength, convectivc heat transfer coefficient, and modulus of elasticity.

3.2.7 Steam chambers. Steam heated chambers may be made of ASTM B 209 aluminum alloy or fabricated steel.

3.2.8 Ironing chest. The ironing chest shall be of ASTM A 36 steel. ASTM A 47 malleable iron, or of steel or iron of equal or better yield strength, convective heat transfer coefficient, and modulus of elasticity. It shall be plated with nickel, chromium, or other corrosion-resisting metals.

3.2.9 Type III-steam cylinder. The type III flatwork ironer steam cylinder shall be of ASTM SA53 grade B steel or of steel of equal or better yield strength, and convective heat transfer coefficient and conforming in all respects to the ASME Boiler and Pressure Vessel Code having ASME stamping and approval.

3.2.10 Type III-auxiliary roll. The type III auxiliary roll shall be of corrosion-resisting steel.

3.2.11 Buck padding. Bucks shall be covered with a preformed corrosion-resisting woven metallic pad which shall be covered with double faced nylon flannel and tailored heat-resisting non-flammable, nylon-polyester blend cover cloth having a draw cord and retaining springs, where needed. The buck, in addition to the above, shall be a non-asbestos insulation material placed between the woven metallic pad and heated buck.

3.2.12 Flexible steam hose assemblies. Flexible steam hose assemblies shall be extruded or convoluted tetrafluoroethylene hose with stainless steel wire braid and end fittings in accordance with SAE AS604 or AS620.

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3.2.13 Insulating materials. Equipment shall be completely free from asbestos or asbestos treated materials, The contractor shall substitute for asbestos only those materials which perform at least as well as asbestos for the intended application.

3.3 Construction.

3.3.1 Safety devices. Exposed belts, chains, shafts, pulleys, gears, and other moving parts shall be fully enclosed and guarded to prevent contact with same. Guards shall be sheet metal or wire mesh with openings so that the fingers of the fifth percentile female in accordance with ASTM 1166 shall not pass through. All parts of thc guard shall be rigid and secured so as to be readily removable without disassembling piper or fittings on the machine

3.3.2 Access to parts requiring frequent adjustment. Provision shall be made in guards, fitted over parts requiring frequent adjustment, to permit access to such parts without the use of special tools (see 3.1.2).

3.3.3 Safety. Laundry pressing and finishing equipment shall reduce to the minimum the possibility of injury to the operators and second party.

3.3.4 Equipment bases. Bases shall be flat on the bottom and lugs for anchor bolts shall be flushed with the bottom of the base or slightly relieved from same, so that the machine will match up to a flat surface and eliminate the need for special foundations. The base supports of machine shall be of steel, ductile iron, or malleable iron and of rigid construction. Machines shall be constructed with provision to permit cleaning underneath the machines where required.

3.3.5 Machine mounting. The laundry press base shall have four or six 5/8-inch bolt holes for anchoring the unit to the deck. Types I and II, classes C, and C-E, presses shall have six bolts 5/8-inch bolt holes and classes B, B-E, G, G-E, N, and N-E presses shall have four or six bolts 5/8 inch bolt holes. The press's mounting configuration shall be in accordance with figure 1. Type III, classes A and B ironers shall have six bolt holes 3/4 inch. The ironer’s mounting configuration shall be in accordance with figures 2 or 3 as specified. Base structure or mounting pads shall be secured to a sufficient area of the unit structure to ensure adherence under ship motion and vibration. Structure penetrated by hold-down bolts shall be solid steel. Feet, legs, and frame shall withstand shipboard forces of 24 and 60 times the acceleration due to gravity in the horizontal and vertical directions respectively about the center of gravity of the unit, without exceeding the yield strength of mounting structure material. No other attachment points shall be permitted except for vent ducting, steam lines, air lines, and electrical cabling.

3.3.6 Heating medium. Type I and type III pressing equipment shall be steam heated. Type II pressing equipment shall be electrically heated.

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3.3.7 Working pressures.

3.3.7.1 Steam working pressure. Pipes, fittings, coils, cylinders. chests, and chambers using steam shall be rated for 100 pounds per square inch (lb/in^2) gauge saturated steam working pressure.

3.3.7.2 Air working pressure. Pipes, fittings, and all parts that arc pneumatically operated shall be rated for a minimum air operating pressure of 65 lb/in^2 gauge. The air system shall show no signs of leaks or distortion (see 4.4.2.10).

3.3.8 Beaings. Bearings shall be selected as recommended by the manufacturer with load ratings including applicable safety factors. Contacting metal swinging surfaces during normal operation, shall be provided with bronze bushings, ball, or roller bearings.

3.3.8.1 Replaceability of bearings. Bearings shall be readily replaceable (see 3.1.2).

3.3.9 Lubrication. Except for motors and where lubrication reservoirs are required, forced lubrication with fittings shall be provided and accessibly located for lubricating chain drives and all bearings (see 3.1.2). Where lubricant reservoirs are required for chain and gear drives, the reservoirs shall provide means for filling, draining, and checking lubricate level. Extended grease fittings shall be provided on machines requiring greasing to be accomplished with the machinery running, to prevent entanglement of maintenance personnel with revolving machinery. Renewable oils and greases shall be of the following types:

a. DOD-G 24508 (high performance, multi purpose grease): intended for ball or roller bearings operating continuously up to 300 degrees Fahrenheit (°F)

b. MIL-L-17331, Military symbol 2190 temperature extra pressure (lubricating oil): intended for general mechanical lubrication

c. MIL-G-23549 (lubricating grease): intended for general purpose

d. MIL-S-8660 (silicone compound): intended to lubricate rubber.

3.3.9.1 Seals. The lubricant shall be prevented from entering parts of the machine where clothes are handled.

3.3.10 Lock nuts. Nuts subjected to shock or vibration shall be locked.

3.3.11 Belt and chain tension adjustment. Tension on belt or chain drives shall be readily adjustable.

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3.3.12 Finish and color. Ironing surfaces shall be polished smooth. Laundry pressing and finishing equipment shall be finished according to the manufacturer's standard practice except for pushbuttons. Emergency-stop and normal-stop pushbuttons shall be red.

3.3.13 Aprons/workthrough, cover cloths, feed ribbons, and pads. Aprons/workthrough, cover cloths, feed ribbons, and pads, where required, shall be furnished complete with machines as applicable.

3.3.14 Piping. Necessary integral piping, fittings, condensate traps, pressure-reducing valves and other valves for air, steam, and drains, shall be furnished complete for installation to supply and drain lines with standard fittings. Interconnecting piping between equipments and to connect equipment with supply and drain lines will be furnished by the installing activity. Piping within and throughout the equipment shall be constructed to accommodate self- and ship-induced vibrations (see 3.7.4).

3.3.14.1 Valves. Steam, drain, air, check, and other operational valves shall be of bronze (see 3.2.3.3). Where required to operate the machine, the valves shall be accessible from the front of the machine. Valves shall be readily accessible for maintenance (see 3.1.2). Steam and air supply connections shall be supplied with valves. Rotary valve controls shall open the valve with a counterclockwise motion. Valve controls shall be provided with double ended arrows showing the direction of operations and labeled at each end to indicate the functional result (for example, open and close). Each valve shall be fitted with a renewable corrosion-resisting disk.

3.3.15 Spacing and extent of fastenings. The spacing and extent of welds, rivets, bolts, and screws shall ensure securing to withstand shipboard vibration (see 3.7.4) and prevent bulging of the metal fastened.

3.3.16 Electrical requirements. Electrical equipment shall conform to the recommended practice for electrical installation aboard ship as specified in MIL-E-917. Power supplied to the equipment shall be 440 volts alternating current (Vac) 60 hertz (Hz) 3 phase. Motors of 1 horsepower size and larger shall not be installed in single phase circuits Electrical equipment shall operate without hazard to personnel or equipment with an accidental ground on any conductor. Control equipment enclosures shall be dripproof in accordance with MIL-STD-108.

3.3.16.1 Motors. Motors shall operate the parent equipment under all operating conditions without exceeding permissible temperature rise or current rating. Motor bearings shall have adequate facilities for lubrication. Motors shall have proper starting operation characteristics for the driven equipment. Motors shall be in accordance with MIL-M-17060 with the following characteristics:

Ambient temperature	–	104 °F
Service	–	C
Enclosure	–	Dripproof
Cooling	–	Natural

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Horsepower	–	As required
Type	–	Squirrel cage induction
Duty	–	Continuous
Voltage	–	440 Vac
Bearing	–	Ball or sleeve
Insulation	–	Class B or F

3.3.16.1.1 Motor wire. Motors shall be wound with wire sizes conforming to the American Wire Gauge (AWG). Half-size wire shall not be used.

3.3.16.2 Controllers. Contactors for power circuits shall switch all three lines in phase circuits and both sides of single phase circuits. Integral motor power circuits shall be three phase and shall be controlled by contractors, which in turn may be actuated by relays. Control circuits shall operate at 115 volts (V) or less. Solenoid valves shall be provided with high temperature epoxy encapsulated coils. Motor controllers shall be magnetic with overload protection relays. Motor controllers shall be provided with overload protection for motors 1/8-horsepower and larger. Reduced voltage control circuits shall be protected by cartridge fuses sized not larger than 200 percent of the transformer kilovoltampere (kVA) rating or 100 percent of the control wire amperage rating, whichever is smaller Electrical components enclosures shall be dripproof in accordance with MIL-STD-108. The control equipment shall have the following characteristics:

Ambient temperature	–	104 °F
Enclosure	–	Dripproof
Operation	–	Manual or magnetic
Type	–	Alternating current (ac): across line
Function	–	To suit equipment
Duty	–	To suit equipment
Performance	–	Manual: nonautomatic
		Magnetic: semi-automatic
Protection	–	Thermal overload and low voltage
		Remote circuit for safety interlock

3.3.16.21 Switches and timers. Each pneumatically and electrically operated unit of pressing and finishing equipment shall be provided with timers, limit switches, remote control pushbutton switches, signals, and indicating lights as necessary to perform the required functions. Pushbutton control stations shall be mounted at points convenient to the operators. Pushbuttons shall be fully protected or recessed to prevent accidental operation (see 3.3.20).

3.3.16.3 Wiring methods and practices. Wiring shall be in accordance with MIL-E-917 and this specification. Wiring between different devices mounted on the machine routed outside of an enclosure shall be in a flexible oil-resistant metal conduit or liquid-tight flexible metal conduit not smaller than 3/8-inch electical trade size. Fitting and junction box connections for conduit entrance shall be either threaded bushings or watertight, neoprene-bushed, stain-relief type. Drainage holes shall be provided at the low point of all conduit runs or junctions. Wiring shall be with moisture-

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resistant thermoplastic insulated copper conductors rated for a minimum operating temperature of 221 °F. Conductors shall be stranded. When wires are run through holes in partitions, the holes shall be equipped with grommets for mechanical protection of insulation. Wire shall not be carried over or bent around any sharp corners or edges. Wire runs in conduit or in harness assemblies shall be marked every 12 inches to correspond with the electrical diagram. Terminal blocks shall be plainly marked to correspond with the markings on the electrical diagram. Solder lugs, soldered joints, and wire-nut type connections shall not be permitted. Electrical connections shall be solderless ring connectors. The ends of each wire shall be connected to terminals on the part or to the terminal boards by means of a pressure or solderless lug terminal. Not more than three connections shall be made at each terminal. Conduit, fittings, and wire shall conform to and be installed in accordance with the National Electric Code. Equipment shall be used on an ungrounded electrical system and no grounds shall be permitted on the machine wiring. Connections shall be made between motors, controllers, timers. pushbuttons, and limiting switches installed on unit. Leads shall be extended from motors and control devices to a connection box and properly identified for connections to wiring beyond the unit. When magnetic controllers are remotely mounted separately from the unit, leads shall be extended from motors and control devices to a connection box and properly identified for connections to wiring beyond the unit. Machines with manual across-the-line starters, when mounted on the unit, shall be provided with wiring between the motor and the starter. Electric cable length shall be kept to a minimum with sufficient use of wire wraps and clamps to prevent wire insulation chaffing. The electrical controls shall be provided with a hinged access door with hardware to tightly secure it closed (key latches and magnetic latches shall not be used). Preventative maintenance tasks shall be easily and safely completed by the fifth to ninety-fifth percentile person and consideration for the use of tools or test equipment shall be taken into account (see 3.1.2).

3.3.16.4 Circuit breakers. Two pole magnetic circuit breakers shall be installed on each transformer primary lead to provide overcurrent protection. The rating of the breaker shall be in accordance with NEMA Section SG-3 requirements. The trip characteristics shall be selected according to the normal and surge current requirements of the transformer supplied circuit. The breaker shall utilize sealed construction.

3.3.17 Tolerances. Tolerances shall be in accordance with ANSI Y14.5M.

3.3.18 Pipe connections to aluminum or aluminum-alloys. Where aluminum or aluminum-alloy is used in presses and finishing equipment, pipe connections shall be fitted with reducing non-ferrous metal bushings.

3.3.19 Drives. Drives shall be as follows:

3.3.19.1 Gear or chain drive. Gears shall be manufactured in accordance with current commercial practice. The type of gearing and the material used shall provide adequate beam strength, resistance to wear, and quietness of operation. Gear mountings, including shafts, bearings

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and supporting structures, and lubrication shall be in accordance with current commercial practice. Gearing, drive chains, and shafting shall be constructed on the basis of the maximum operating torque of the motor. Gears and chains shall be readily accessible for inspection and means shall be provided for adjustment of tension (see 3.1.2).

3.3.19.2 V-belt. Belts shall be V-type provided in accordance with the belt manufacturer's recommendations based on load ratings, with adequate safety factors. The V-belts shall have a minimum service life of 5000 hours of machine operation. Pulley diameters shall be not less than recommended by the belt manufacturer. V-belts shall be readily accessible for inspection and means shall be provided for adjustment of tension (see 3.1.2).

3.3.20 Controls, gauges, indicator lights, and audible signals.

3.3.20.1 General criteria. Controls, gauges, indicator lights, and audible signals shall be selected and integrated into the laundry presses and finishing equipment to meet the applicable human factors engineering criteria of ASTM 1166-88 and herein.

3.3.20.1.1 Distribution of work load. Controls shall be selected and distributed so that none of the operator's limbs will be overburdened. Equipment design shall take into account the following considerations:

a. The nature, frequency, and difficulty of the related tasks to be performed by the operator

b. The position of the body during the performance of tasks

c. Mobility or flexibility requirements imposed by these tasks.

For further information on arthropometrie data see ASTM 1166-88.

3.3.20.1.2 Stops. Controls which should not be operated past their start/end positions or limits shall be provided with stops to prevent such movement.

3.3.20.1.3 Consistency of movement. Direction of control movement shall be consistent with the related movement of an associated display or equipment component. Movement of a control forward, clockwise, to the right, up, or pressing a control, shall turn the equipment or component on, or cause the quantity to increase.

3.3.20.1.4 Grouping. Controls which function in sequential operation necessary to a particular task, or which operate together, shall be grouped together along with their associated displays. When several steps of a sequence are selected by one control, the steps shall be arranged by order of occurrence to minimize control movements and prevent cycling through unnecessary steps. Cycling through the control's ON/OFF position shall be avoided.

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3.3.20.1.5 Sequential operation. Where sequential operations follow a fixed pattern, controls shall be arranged to facilitate operation (for example, in a pattern left to right and top to bottom, as a printed page).

3.3.20.1.6 Location of primary controls. The most important and frequently used controls shall have the most favorable position with respect to ease of reaching and grasping (particularly rotary controls and those requiring fine settings). When a transilluminated indicator is associated with a control, the indicating light shall be so located that it can be associated with the control without error and shall be visible to the operator during control operation.

3.3.20.1.7 Spacing. Minimum spacing between a control and any adjacent obstruction shall be in accordance with ASTM 1166. The spacing between adjacent edges of simple round indicator light fixtures shall permit unambiguous labeling, signal interpretation, and bulb removal.

3.3.20.1.8 Control colors. Controls shall be black or grey. If color coding is required, only the following colors shall be selected for color coing: red, green, orange-yellow, white, and blue. Blue shall only be used if an additional color is absolutely necessary.

3.3.20.1.9 Relation to display. When color coding is used to relate a control to its corresponding display, the same color shall be used for both control and display. Color coding shall be in accordance with ASTM 1166.

3.3.20.1.10 Control panel contrast. The color of the control shall provide contrast between the panel background and the control.

3.3.20.1.11 Coding. Indicator lights and controls shall be coded in accordance with ASTM 1166. Different sizes are intended only for the attention-getting value which larger lights of at least equal luminance provide in relation to indicator lights of lesser importance. No more than three different size lights and controls shall be used for discriminating importance by size.

3.3.20.1.12 Prevention of accidental actuation. Controls shall be configured and located so that they are not susceptible to being moved accidentally (for example controls with resistance such as viscous or Coulomb friction, spring-loading. controls that are recessed, and so forth).

3.3.20.1.13 Labeling of controls. Controls and displays shall be appropriately and clearly labeled with label plates, with the basic information needed for proper identification, utilization, actuation, or manipulation of the element, except where it is obvious to the operator what an item is and what he or she is to do with it. The labels shall convey verbal meaning in the most direct manner by using simple words or phrases. Each control or display shall be labeled according to function, and the following criteria shall apply:

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a. Similar names for different controls and displays shall be avoided.

b. Instruments shall be labeled in terms of what is being measured or controlled, taking into account the user and purpose.

c. Control labeling shall indicate the functional result of control movement (for example, on, off, and so forth). Such information shall be visible during normal operation of the control.

d. Labels shall be placed above the controls or displays they describe and oriented horizontally so that they may be read from left to right. When the panel is above eye level, labels may be located below if the label visibility will be enhanced.

e. The units of measurement (for example, °F, minutes, and so forth) shall be labeled on the panel.

f. Lettering shall be dark on a light background or vice versa.

3.4 Identification plate. Identification plates shall be made for the machine and its major components that is, boilers, electric motors, and so forth. Identification plates shall be made of corrosion -resisting steel,, aluminum alloy or brass. Characters shall be not less than 3/16 inch in height. Characters shall be stamped on the plate. Stamping shall be not less than 0.003 inch deep. The thickness of the plate shall be 0.03 inch minimum. Plate information shall remain permanent and legible for the life of the equipment.

3.4.1 Mounting. Each identification plate shall be securely affixed to each machine in a conspicuous place.

3.4.2 Identification plate data. Data marked on each identification plate shall include the following:

a. Manufacturer's type or model number, serial number, contract or order number, and date of manufacture

b. Salient design features including size, voltage, current, and working pressures

c. Name of machine and stock number

d. Component identification number. The component identification number entry shall consist of the letters "CID", followed by a 9-digit number. No unit shall be shipped without the CID number and its associated components shall also be recorded on the packing list.

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3.5 Press dimensions type I and II. Laundry presses shall have an unpadded buck dimension as specified in table I. Dimensions shown are approximate: length tolerance shall not exceed plus or minus 1 inch, width tolerance shall not exceed plus or minus 1/2 inch, and radial tolerance shall not exceed 1/4 inch.

TABLE I. Buck dimensions.

Class or type		Length (max) inches	Width (max) end inches	Radius (max) end inches	Width (ctr) inches	Width (min) end inches	Radius (min) end inches	Pressing requirements
Electric	Steam							
B-E	B	35 1/2	--	-	12	7 both	-	Trouser topping (double) two at a time
C-E	C	53 1/4	18	51 1/2	-	10 1/2	34 1/2	General utility
G-E	G	2 bucks, one for the collar and yoke and one for the cuffs						Collar and cuffs yoke of shirts in two lays
N-E	N	Two tandem bucks each (approximately) 22 inches long by 11-inches wide with a 5-1/2 inch radius outer ends; bucks to be spaced 6 1/2 inches apart						Shirt sleeves both in one lay; noncabinet

3.5.1 Overall dimensions. Overall dimensions of the table or framework including operating mechanism, handles and pedals, for supporting each specified press, shall not exceed the dimensions shown in table II. Type I and II presses shall be such that they may be disassembled, transported, and reassembled within the ship without hot work, such as cutting, welding, and brazing (see figure 4). Major subassemblies shall bolt together. Limiting dimensions for each major subassembly shall be 50 inches high, by 25 inches wide, by 36 inches deep. Subassemblies shall be matched-marked to assist in field assembly.

TABLE II. Dimensions of table or framework.

Class or type		Width (inches)	Depth (inches)
Steam	Electric		
B	B-E	49	44
C	C-E	61	44
G	G-E	49	44
N	N-E	54	46

3.5.1.1 Press head configuration. Press heads shall be unpadded and of a size and configuration to mesh with and completely cover the padded bucks when closed.

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3.5.2 Details of units or parts.

3.5.2.1 Type I-laundry presses, steam heated. Each press shall consist of a rigid steel frame supporting a steam heated head or heads and a steam heated buck or bucks. Where multiple heads or bucks are required, the heads shall be properly mounted to operate simultaneously from one yoke. The frame shall also support the work table, operating mechanism, and the operating mechanism enclosure. The operating mechanism shall be enclosed to the maximum extent practicable, the enclosure being fitted with access plates neccessary for inspection and maintenance of the operating mechanism (see 3.1.2). Strength members and operating parts shall be steel or malleable iron (see 3.2.5).

3.5.2.1.1 Head design. The upper or top surface of press heads shall be ASTM A 36 steel or ASTM A 47 malleable iron. The ironing or lower surface shall be smooth and either polished or plated. The head shall be concave in shape to conform with the buck. Press heads shall conform to the ASME Boiler and Pressure Vessel Code, section VIII (see 3.2.6).

3.5.2.1.2 Buck design. The buck shall be constructed of steel, ductile iron, or malleable iron and rigidly mounted and connected to the frame. The top plate of the buck shall be curved and shaped to fit the head and the articles for which the press is especially designed for. The bottom plate of the buck shall be made of the same material as the upper plate. Bucks shall conform to the ASME Boiler and Pressure Vessel Code, section VIII (see 3.2.6).

3.5.2.1.3 Buck padding. A resilient pad with additional padding and covering shall ensure uniform pressure and conform to the uneven thickness of goods being pressed. The padding and covering shall include a heavy knitted cotton pad, double-faced cotton flannel or table felt, and tailored cotton cover cloth complete with draw cord and retaining springs.

3.5.21.4 Head and buck arrangements. The head shall be movable and spring counterbalanced when required for ease of movement. Radial movement and alignment of the head shall offer no interference or danger to the operator in the performance of pressing functions and shall be positioned for proper contact with the buck.

3.5.2.1.5 Head and buck adjustment. Means shall be provided for automatically adjusting either the head or buck, or both head and buck, to conform to varying thicknesses of goods being pressed.

3.5.2.1.6 Table. A finished, rigid metal work table, shall be furnished with each press. Table and edge aprons shall support press work that overlaps the buck and shall be snag free.

3.5.2.1.7 Steam and condensate. Each head and buck shall be uniformly heated and properly drained. The steam and condensate lines to and from moving members shall be connected by means of flexible metallic hose rated for a saturated steam working pressure of 100 lb/in^2 gauge.

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3.5.2.1.8 Pneumatic power application. Presses shall be arranged so that the head or heads shall be moved automatically to the full open position when the pneumatic power is off. The head or heads shall be moved continuously and uniformly by air power only, so as to make contact with the opposite member. In this position, air power shall supply the necessary pressure for pressing materials. The amount of pressure shall be adjustable. The press head and buck shall provide automatic takeup to compensate for shrinkage of padding. The press shall ensure satisfactory pressure for garments of unequal thickness, as well as those of uniform thickness. In case of an obstruction between the head and padded buck amounting to 3/4 inch or more, the moving member shall not lock closed and upon release of one or more of the closing bottons, the head shall release to the open position. Presses supplied for shipboard use shall have the same basic operating characteristics and the same pressures between heads and bucks as those supplied for commercial use. Presses shall be pneumatically operated and shall operate on a minimum working air pressure of 65 lb/in^2 gauge.

3.5.2.1.9 Press controls. Power for the moving member or members shall be controlled by two manually operated controls located not less than 10 inches apart, recessed, and arranged so that both controls must be operated simultaneously, thereby requiring the use of both hands of the operator during the entire closing and locking operation. Upon releasing one or both controls before the head has reached the closed position, the moving member shall return automatically and promptly to the open position. Personnel between the fifth and ninety-fifth percentiles of anthropometric dimensions shall activate the press without bending, leaning, or excessive reaching (see 3.1.2 and 3.3.20). After pressure has been applied, operation of one or both controls or of a separate manual control shall return the moving member or members to the open position.

3.5.2.1.10 Utility hook-ups. The datum point is located at the deckplate about the front left bolt center with the X axis parallel to the front edge of the machine pointing to the right, the Y axis parallel to the side edge of the machine pointing to the rear, and the Z axis pointing up. Utility locations with respect to the datum point shall a maximum tolerance limit of plus or minus 3/8 inch in each given plane. The utility hook-up locations relative to the datum point are:

		X direction (inches)	Y direction (inches)	Z direction (inches)
a.	Steam inlet 3/8 inch (female - national pipe thread (FEM NPT))	18	27 1/2	27
b.	Steam outlet 3/8 inch (FEM NPT)	0	27 1/2	25 1/2
c.	Air inlet 3/8 inch (FEM NPT)	(right rear corner).		

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3.5.2.2 Type II laundry presses, electrically heated. Each press shall consist of a rigid frame supporting an electrically heated head and buck. Where multiple heads or bucks are required, the head shall be properly mounted to operate simultaneosly from one yoke. The frame shall also support the work table, operating mechanism, and the operating mechanism enclosure. The operating mechanism shall be enclosed to the maximum extent practicable, the enclosure being fitted with access plates, as necessary, for inspection and maintenance of the operating mechanism (see 3.1.2). Strength members and operating parts shall be constructed of steel or malleable iron (see 3.2.5).

3.5.2.2.1 Press heads. The top surface of press heads shall be covered with sheet metal. The lower surface shall be smooth and either polished or plated (see 3.2.6).

3.5.2.2.2 Press bucks. Bucks shall be rigidly mounted on the press frame. The top plate of the buck shall be curved and shaped to fit the head and configured to efficiently press the intended articles of clothing (see 3.2.6).

3.5.2.2.3 Electric heating of heads and bucks. Press heads and bucks shall be heated by sheathed, electrical strip heating elements. The electric strip heating elements shall be designed into the head and buck to allow replacement of damaged elements by shipboard maintenance personnel. Elements shall be configured and arranged for efficient and uniform heating of each pressing surface. The arrangement of temperature sensors shall preclude overheating of any area in event of element failure. The thermostat controller for the press heads shall be provided with a maximum temperature setting of 320 °F. The thermostat controller for the press bucks shall be provided with one temperature setting of 275 °F. The thermostats shall be provided with an indicator light to depict energization of the electric strip heating elements in the head and buck. Sheath material shall be corrosion-resistant with a minimum sheath material operating temperature of 900 °F. The sheath shall enclose a 80Ni-20Cr helically coiled resistance wire uniformly spaced over the length and width of the element. Mounting slots or holes shall be provided at both ends of the heating elements. Shielding of element threaded terminals and wiring shall eliminate electric shock, moisture infiltration, or burn hazards for operators and maintenance personnel. Bending or forming shall be done at the element manufacturer's plant. See 4.4.2.4 through 4.4.2.4.6 for testing of heating elements and for testing the temperature distribution across the head and buck.

3.5.2.24 Utility hook-ups. The datum point is located at the deckplate about the front left bolt center with the X axis parallel to the front edge of the machine pointing to the right, the Y axis parallel to the side edge of the machine pointing to the rear, and the Z axis pointing up. The utility hook-up locations relative to the datum point are:

a. Electric cable: left rear corner
b. Air inlet 3/8 inch (FEM NPT) right rear corner

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3.5.2.3 Type III, flatwork ironer. The flatwork ironer shall be of the heated cylinder type, with feed ribbons compression roller, and return ribbons arranged to carry the flatwork type laundry from the feed side to the heated cylinder and back to the receiving table on the feed side. The flatwork shall be kept in contact with approximately 75 percent of the heated cylinder by a padded roller and a set of continuous ribbons.

3.5.2.3.1 Ironing rate. The ironer shall be configured so that nearly the entire length of the heated cylinder will come in contact with the flatwork being ironed. Rate shall be such that sheets containing not more than 45 percent of their dry weight in moisture will iron dry when the machine is operated at a normal speed of not less than 14 feet per minute and steam working pressure is mainlined at 100 lb/in gauge. The ironing feed rates shall be adjustable from 16 ± 2 feet per minute to 36 ± 2 feet per minute.

3.5.2.3.2 Steam cylinder. The cylinder shall be of ASTM SA53 grade B steel, shall conform to the ASME Boiler and Vessel Pressure Code, section VIII for unfired pressure vessels, (see 3.2.9) and shall have suitable means for draining. Limiting dimensions of steam cylinder shall be as specified in table III.

TABLE III. *Steam cylinder dimensions.*

Class	Approximate outside diameter (inches)	Minimum length (inches)	Maximum length (inches)
A	18	90	100
B	18	65	75

3.5.2.3.3 Steam working pressure. The cylinder and all working parts connected there with shall be constructed and rated for a saturated steam working pressure of 100 lb/in^2 gauge minimum to 125 lb/in^2 gauge maximum.

3.5.2.3.4 Auxiliary rolls. There shall be one padded compression roller having a diameter of approximately 6-3/8 inches when padded. The roll shall be of steel with heads and trunnions rigidly secured. The padding of the compression roller, cover cloth and return ribbons shall be made of temperature resistant fabrics. A lever shall be furnished to raise and lower the padded compression roller from contact with the heated cylinder. A padded finger roll (doffer roll) shall be mounted on the feed ribbons in front of the finger guard safety device. This roll shall act as a guide in feeding and shall eliminate wrinkles from the work.

3.5.2.3.5 Ribbons. The ribbons shall be of duck, nomex, or similar material of equivalent characteristics. The material shall be fitted to prevent the flatwork from winding around the auxiliary rolls and to deliver the pressed flatwork back to the receiving table.

3.5.2.3.6 Work trough. An apron or work trough of canvas or equivalent material shall be suspended across the front of the machine at a height for holding flatwork while the leading edge is being fed into the flatwork ironer.

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3.5.2.3.7 Alignment guide. Means shall be provided to automatically hold the feed ribbons and the return ribbons in tension and alignment.

3.5.2.3.8 Bearings. The rolls and cylinder shall be fitted with self-aligning ball or roller bearings.

3.5.2.3.9 Frame. The flatwork ironer shall have a frame for supporting the steam heated cylinder, rolls, drive mechanism, and attached accessories. Frame shall be rigidly constructed and braced. Overall dimensions of the flatwork ironer shall be as specified in table IV.

TABLE IV. Flatwork ironer's overall dimensions.

Class	Maximum length diameter (inches)	Maximum depth (inches)	Maximum height (inches)
A	130	54	70
B	105	54	70

3.5.2.3.10 Anchor bolts. The frame shall be provisioned for six 3/4-inch nominal diameter anchor bolts configured to the dimensions shown on figure 2 for class A ironers and figure 3 for class B ironers.

3.5.2.3.11 Receiving table. A rigid corrosion-resisting metal table shall be fitted along the feed side (front) of the machine at the proper height for receiving ironed flatwork from the machine. The table shall hold the flatwork and shall be snag free.

3.5.2.3.12 Safety guard. A guard of aluminum alloy or corrosion-resisting metal shall be fitted across the entire line of feed, arranged so that a slight movement of the guard in the feeding direction shall immediately stop the machine.

3.5.2.3.13 Drive mechanism. The machine shall be driven by an electric motor, mounted on the frame of the machine, and connected by gearing. V-belts, or chains. The motor shall be of adequate horsepower to drive the machine continuously at the required rates (see 3.3.16.1 and 3.5.2.3.1).

3.5.2.3.14 Controls. The machine shall be furnished with all necessary controls and wiring (see 3.3.16 through 3.3.16.4 and 3.3.20).

3.5.2.3.15 Dismantling. The ironer shall be shipped dismantled when specified (see 6.2.1) so that the component parts can pass through a 26- by 66-inch ship type door (see figure 4). Subassemblies shall be matched marked to assist in field assembly.

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3.5.2.3.16 Utility hook-ups. The datum point is located at the deckplate about the front left bolt center with the X axis parallel to the front edge of the machine pointing to the right, the Y axis parallel to the side edge of the machine pointing to the rear, and the Z axis pointing up. The utility hook-up locations relative to the datum point are:

	X direction (inches)	Y direction (inches)	Z direction (inches)
85-inch ironer			
a. Steam inlet 1 inch FEM NPT	(left rear corner)		
b. Steam outlet 3/4 inch FEM NPT	(left rear corner)		
c. Electric cable	(right rear corner)		
60-inch ironer			
a. Steam inlet 1 inch FEM NPT	(left rear corner)		
b. Steam outlet 3/4 inch FEM NPT	(left rear corner)		
c. Electric cable	(right rear corner).		

3.6 Performance.

3.6.1 Press operation. Presses supplied for shipboard use shall have the same basic operating characteristic as those supplied for commercial use, with 600 pounds pressure between heads and bucks. Presses shall be pneumatically operated.

3.6.2 Working pressures.

3.6.2.1 Saturated steam. Pipes, fittings, coils, cylinders, chests, and chambers using steam shall be rated for 100 lb/in^2 gauge saturated steam working pressure. The steam system shall show no signs of leaks or distortion (see 4.4.2.8).

3.6.2.2 Compressed air. Pipes, fittings, and all parts that are pneumatically operated shall be rated for a minimum air operating pressure of 65 lb/in^2 gauge (see 4.4.2.9 and 4.5.3.3).

3.6.3 Inclined operation. Equipment shall function in accordance with the requirements of this specification, without reduction in performance, when ship is rolling 12 degrees starboard in a 4-1/2 second period (see 4.4.2.1).

3.6.4 Vibration. Equipment shall operate normally while being subjected to vibration levels from 4 to 25 Hz in accordance with MIL-STD-167-1, type I environmental equipment (see 4.4.2.2). The equipment shall perform its function during and after the testing. Minor damage or distortion will be permitted during testing providing the equipment performs its principle functions. Major

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failures are considered to be those that would cause malfunction of the equipment or present a safety hazard. Nonrepetitive failures of such parts as wiring and solenoid valves, which can be easily replaced or repaired, arc generally considered minor failures.

3.65 Electromagnetic compatibility (EMC) requirements The equipment shall be tested to the applicable EMI requirements of MIL-STD-461, class A4, with the following modification:

a. The equipment shall be tested for collection of CE01, CE03, and RE02 electromagnetic interference interface data in accordance with MIL-STD-461 (see 4.4.2.3).

3.6.6 **Shipboard power.** Electrically powered equipment shall meet the following shipboard performance requirements (see 4.4.2.10).

Test	Performance criteria
Steady state voltage	Unit shall operate during each test condition
Power interruption	Unit shall return to normal operation without deterioration in performance when the power is restored.
Leakage current	Unit under normal operation shall not have a leakage current greater than 5 milliamperes (mA).
Power factor	The unit's overall power factor shall be in the range of 0.8 lagging to 0.95 leading under average voltage tolerance conditions.
Load unbalance	The unit's kVA unbalance during normal operations shall not exceed 5 percent.
Dielectric strength	The unit's circuits shall experience no disruptive discharge or deterioration
Insulation resistance	The units electrical components shall have a minimum cold insulation of 10 megohms.

3.6.7 **Electric heating element strength.** The heating elements shall meet the requirements specified herein (see 4.4.2.4.1) and shall not short out (see 4.4.2.4.2) The hot and cold resistance shall not exceed a resistance of 10 megohms (see 4.4.2.4.3) and 25 megohms (see 4.4.2.4.4) The heating elements shall be within plus or minus 5 percent of the rated watt density per unit length (see 4.4.2.4.6).

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3.6.8 Electric motors. Each electric motor shall adequately perform when subjected to its maximum operational load. The electric motor shall be within 5 percent plus or minus its required voltage and shall not exceed its maximum rated current draw (see 4.4.2.7).

3.6.9 Operation. Each equipment type and class shall perform its operations (see 4.4.2.6).

3.6.10 Enclosures. Equipment with electrical control panels or electric motors shall operate satisfactorily and have no accumulation of water within the enclosure (see 4.4.2.5).

3.6.11 Finished painted surface requirements

3.6.11.1 Salt spray requirements. There shall be no evidence of blistering of the items finished painted surface after 240 hours of exposure in the salt spray test specified in ASTM B 117. Creepage, or loss of adhesion, shall be measured at the scribed marks after scrapping. This creepage shall not extend more than 1/8 inch on either side of scribed marks. The total width of gap in the prime coat shall not exceed 1/4 inch (see 4.4.2.11.1).

3.6.11.2 Hardness requirements. The item’s finished painted surface shall remain undamaged when subjected in the test specified in 4.4.2.11.2.

3.6.11.3 Adhesion requirements. A classification of not more than one panel out of two shall be allowed when item's finished painted surface is subjected to the test specified in 4.4.2.11.3.

3.6.12 Ironing rate, flatwork ironer. The ironing rate shall be such that sheets, containing not more than 45 percent of their dry weight in moisture, will iron dry when the ironer is operated at a normal speed of not less than 14 feet per minute with the steam pressure maintained at 100 lb/in^2 gauge.

3.6.13 Distribution of temperature. Head and bucks for type II presses shall be within an average temperature of plus or minus 10 °F of the thermostat setpoint with a maximum standard deviation of 15 °F. The surface temperature limits on the buck and head shall be within plus or minus 40 °F of the thermostat setpoint. The buck temperature shall be 275 °F. The head shall have a maximum temperature setting of 320 °F (see 4.4.2.4.5).

3.7 Workmanship. The finished end items shall be free from defects such as fractures, splits, punctures, tears. dents. creases, bows, miscasts, deterioration, or malformation and as follows:

a. Welds shall be continuous (on structural supports), sound, smooth, and free from porosity, cracks, incomplete fusion, and deformation of material. All scale and flux (when flux is used) shall be removed from the welds.

b. Threaded fasteners shall not be broken, stripped, bent, fractured, loose, or missing.

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c. Rivets shall be driven completely to fill holes. The rivet heads shall be in full contact with the surfaces of the riveted member.

d. Wiring shall not be cut, abraded, or have excess insulation stripped. Wiring shall be tightly joined at the terminals. Wiring harnesses shall be secured using clamps and/or ty-wraps to prevent chaffing. Wiring shall not be bent or carried through on sharp protruding edges.

e. Surfaces shall be free from burrs, sharp edges, or slivers.

f. Bearings, shafts, and gears shall be properly aligned.

g. Safety guards shall be securely fastened to the equipment.

h. Pressure vessels shall be ASME stamped and approved.

l. The finish applied to the end item shall be continuous, smooth, adherent, without discolorations or foreign materiel embedded, and contains no sags, runs, drips, creeps, laps bubbles, streaks, wrinkles, blisters, cracks, scratches pores pits lumps flux or orange peel. No rust, rough grinds, or toolmarks shall show through the coatings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies anti services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5 The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence or any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inpection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections shall be performed in accordance with test conditions specified herein.

4.3.1 Intermediate visual and dimensional examination. The intermediate product or assemblies which cannot be checked or measured after final assembly shall be examined at various stages of manufacture for conformance to this specification. Examination shall be made at an appropriate time for defects in construction, workmanship, dimensions, and material. Any deviation from specified requirements shall be classified as a defect. The inspection level in accordance with MIL-STD-105, shall be S-1 with an acceptable quality level (AQL) of 4.0 defects per hundred units.

4.4 First article inspection. The first article inspection shall consist of the examinations shown in table V and the tests specified in 4.4.2 (see 6.3).

TABLE V. Classification of defects.

Examine	Defect	Classification	
		major	minor
Materials	Components or parts not fabricated of specified materials (see 3.2.1 through 3.2.13)	X	
Design/general	Guards not as specified (see 3.3.1 and 3.3.2)	X	
	Bases and mounting not as specified (see 3.3.4 and 3.3.5)	X	
	Bearings not as specified (see 3.3.8, 3.3.8.1 and 3.6.2.3.8)	X	
	Lubrication reservoirs, seals, and fittings not as specified (see 3.3.9 and 3.3.9.1)	X	
	Nuts not locked as specified (see 3.3.10)		X
	Belts and chains not as specified (see 3.3.11, 3.3.19.1 and 3.3.19.2)	X	
	Piping and valves not as specified (see 3.3.14 and 3.3.14.1)	X	
	Motors not as specified (see 3.3.16, 3.3.16.1, 3.3.16.1.1, and 3.5.2.3.13)	X	

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TABLE V. *Classification of defects* – *Continued.*

Examine	Defect	Classification	
		major	minor
	Controllers not as specified (see 3.3.16.2)	X	
	Wiring not as specified (see 3.3.16.3)	X	
Details of units or parts	Buck dimensions not as specified (see 3.5)	X	
	Overall dimensions not as specified (see 3.5.1 and 3.5.2.3.9)	X	
	Air system not as specified (see 3.5.2.1.8)	X	
	Steam system not as specified (see 3.5.2.1.7 or 3.5.2.3.3)	X	
	Head not as specified (see 3.5.1.1, 3.5.2.1.1 or 3.5.2.2.1, 3.5.2.1.4 and 3.5.2.1.5)	X	
	Buck not as specified (see 3.5.2.1.2 or 3.5.2.2.2, 3.5.2.1.3, 3.5.2.1.4 and 3.5.2.1.5)	X	
	Table not as specified (see 3.5.2.1.6)	X	
	Press controls not as specified (see 3.5.2.1.9)	X	
	Heating elements not as specified (see 3.5.2.2.3)	X	
	Steam cylinder not as specified (see 3.5.2.3.2)	X	
	Auxiliary rolls not as specified (see 3.6.5.3.4)	X	
	Ribbons not as specified (see 3.5.2.3.5)	X	
	Workthrough/apron not as specified (see 3.5.2.3.6)	X	

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TABLE V. Classification of defects - Continued.

Examine	Defect	Classification	
		major	minor
	Receiving table not as specified (see 3.5.2.3.6)	X	
	Safety guard not as specified (see 3.5.2.3.12)	X	
	Dismantling not as specified (see 3.5.1 and 3.5.2.3.15)	X	
	Utility locations not as specified (see 3.5.2.1.10, 3.5.2.2.4, or 3.5.2.3.16)	X	
Identification plate	Plate material not as specified (see 3.4)	X	
	Plate not mounted securely (see 3.4.1)	X	
	Plate not conspicuous (see 3.4.1)		X
	Plate data not as specified (see 3.4.2)	X	
Workmanship	Welds not as specified (see 3.7 (a))	X	
	Threaded fasteners not as specified (see 3.7 (b))	X	
	Rivets not as specified (see 3.7 (c))	X	
	Wiring not as specified (see 3.7 (d))	X	
	Surfaces not as specified (see 3.7 (e))	X	
	Bearing, shafts, and gears not as specified (see 3.7 (f))	X	
	Safety guards not as specified (see 3.7 (g))	X	
	Pressure Vessels not as specified (see 3.7 (h))	X	
	Finish not as specified (see 3.7 (i))		X

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4.4.1 First article sample. Before the start of production, one machine of each type and class shall be submitted for the tests and examinations specified in 4.4.2.

4.4.2 Tests. Tests shall be as follows:

4.4.2.1 Inclination Type I, II, and III machines shall be inclined at an angle of 12 degrees to the left, right, front, and back. Unit shall operate in each of the four positions for 30 minutes (see 3.7.3).

4.4.2.2 Vibration. The test unit, under normal operating conditions, shall be tested in accordance with MIL-STD-167-1, type I equipment in the range of 4 to 25 Hz (see 3.6.4).

4.4.2.3 Electromagnetic interference. MIL-STD-461, class 4A, tests CEO1, CE03, and RE02 shall be performed on each machine. Noise emissions shall be plotted with the graphical limit emissions specified in MIL-STD-461 for surface ships. Test set-ups are specified in MIL-STD-462 (see 3.6.5).

4.4.2.4 Electric heating element tests for type II presses.

4.4.2.4.1 Voltage. Heating elements shall be subjected to 105 percent of the operating voltage for a period of 10 minutes and shall operate in accordance with the requirements specified herein.

4.4.2.4.2 Dielectric test. Laundry pressing and finishing equipment shall be subjected for 1 minute to a dielectric test voltage. The effective potential, which is twice the rated voltage plus 1,000 volts, shall be applied between points of polarity. The frequency of the test voltage shall be 60 Hz, root mean square (rms), ac, and shall approximate a true sine wave (see 3.6.7).

4.4.2.4.3 Insulation resistance, hot. The insulation resistance shall be measured with a 500 volts direct current (Vdc) insulation resistance tester 5 seconds after operating of the element at the maximum operation temperature for the sheath material, end seal, or element wire; whichever is largest, to determine conformance to 3.6.7. To prevent the heating element from being cooled before the test, the ambient fluid shall not be circulated after the power is removed.

4.4.2.4.4 Insulation resistance, cold. The insulation resistance shall be measured with a 500 Vdc insulation tester with the electric heating elements at room temperature. The cold insulation resistance shall be a minimum of 25 megohms (see 3.6.7).

4.4.2.4.5 Temperature distribution test. On the head pressing surface, straight lines shall be drawn parallel to and 1 inch m from each edge. The left and right ends shall be divided into four equal segments. Three straight lines shall be drawn lengthwise across the head interconnecting the segment division marks. Front to back lines perpendicular to the lengthwise centerline shall be drawn at 4-inch internals. Temperature readings shall be taken at the intersection of the drawn lines. On the buck pressing surface, two horizontal lines 2 inches apart about the centerline shall be drawn. Using these two horizontal lines as references, 2-inch grid squares shall be laid out.

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The temperature shall be measured in the center of every other square. Partial squares with less than half the area of a full square shall be disregarded. A handheld surface temperature probe shall be used to measure the temperature of each designated point. The head thermostat temperature shall be set at 320 °F and the buck thermostat at 275 °F. The press shall be allowed to operate for 1 full hour or until each thermostat cycles to control the temperature three times, whichever is longer. Two temperature readings shall be taken at each designated point; one when it cycles on and one when it cycles off. Measurements shall be taken in groups consisting of one complete row of points. The elapsed time to measure all the points in each row shall be recorded. When the measurements for each row are completed, the temperature of the first point shall be remeasured and recorded. The press shall remain in the open position for the duration of the test (see 3.6.13).

Calculations:

a. Average temperature = $T = 1/N \sum_{i=1}^{N} T_i$

b. Standard deviation = $r = \left(\sum_{i=1}^{N} ((T_i)^2 - N \times (\bar{T}^2))/ N \right)^{1/2}$

4.4.2.4.6 Wattage. The wattage of the heating elements shall be measured to determine conformance to 3.6.7.

4.4.2.5 Dripproof test. Equipment containing electrical controls or motors shall undergo the following dripproof test. The minimum amount of water shall be 1000 cubic inches per square foot of area covered by the drip or sprinkle in 60 minutes. The dripping or sprinkle rate and distribution shall be approximately uniform. The enclosed equipment shall be oriented in five different positions during the test, normal position and inclined at 15 degrees forward, backward, and to each side. The minimum time in each position shall be 12 minutes (8 minutes with the equipment operating and 4 minutes off) (see 3.6.10).

4.4.2.6 Functional test. Each machine shall be checked and tested to determine conformance to the following requirements:

Type I and II presses

a. Two button/hand head closure (see 3.6.2.1.9)
b. Automatic adjustment to varying thicknesses of goods being pressed (see 3.6.2.1.5)
c. Non-1ocking head closure on an obstruction greater than 3/4 inch (see 3.6.2.1.8)
d. Press shall press the garments for 10 hours.

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Type III ironers

a. Adjustment of feedrates (see 3.6.2.3.1)
b. Ironing rate satisfactory (see 3.6.2.3.1)
c. Engaging and disengaging compression roller contact with the heated cylinder (see 3.6.2.3.4)
d. Finger roll guard deactivates machine when moved (see 3.6.2.3.12)
e. Finger roll guides flatwork into ironer (see 3.6.2.3.4)
f. Receiving table at the proper height to receive ironed flatwork (see 3.6.2.3.11)
g. Ironer shall provide finished flatwork for 10 hours.

4.4.2.7 Power input. Motors shall not draw in excess of rated current after accelerating to full speed (see 3.6.8).

4.4.2.8 Steam coils. Steam coils shall be tested hydrostatically at 150 lb/in^2 gauge for 30 minutes with no indication of leaks or distortion (see 3.6.2.1).

4.4.2.9 Pneumatic assemblies. Pneumatic assemblies shall be tested at 130 lb/in^2 gauge static an pressure for 30 minutes without indication of leaks or distortion (see 3.6.2.2).

4.4.2.10 Electrical tests. Equipment supplied with electricity shall meet the requirements specified in 3.6.6.

4.4.2.10.1 Steady state voltage and frequency. The equipment shall be operated for at least 15 minutes at the following three conditions:

a. 440 Vac supplied equipment

(1) 440 Vat, 60 Hz
(2) 418 Vat, 60 Hz
(3) 462 Vat, 60 Hz.

4.4.2.10.2 Power interruption. With the equipment operating within the steady state voltage and frequency tolerances specified in 4.4.2.10.1. the external power supply shall be suddenly interrupted. After an interval of 0.5 to 20 seconds. the power supply within the steady state tolerances shall be reapplied and the equipment restarted. The test shall be repeated four times.

4.4.2.10.3 Leakage current. With the frame of the equipment ungrounded and the equipment operating within the steady state voltage and frequency tolerances specified in 4.4.2.10.1. the leakage current for each phase between the frame shall be successively measured through a 1500 ohm resistive load.

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4.4.2.10.4 Power factor. With the equipment operating within the steady state voltage and frequency tolerances specified in 4.4.2.10.1, the voltage, current. and wattage shall be measured and the power factor computed:

3 phase power factor = wattage/(average voltage x average current $x\sqrt{3}$)

4.4.2.10.5 Load unbalance. With the equipment operating at 440 Vac, 60 Hz, the current on each phase shall be measured and the load unbalance computed:

Load unbalance = (maximum current – minimum current)/(average current)

4.4.2.10.6 Dielectric strength test. A dielectric test voltage of twice the rated voltage plus 1000 Vat, 60 Hz having a capacity of at least 1 kilowatt (kW) shall be applied successively between each phaseline and the equipment frame. The voltage shall be raised gradually and shall be maintained for 1 minute plus or minus 5 seconds.

4.4.2.10.7 Insulation resistance. The insulation resistance between all mutually insulated points and between all insulated points and the frame shall be measured with a 500 Vdc megohmeter after the dielectric test specified in 4.4.2.10.6.

4.4.2.11 Painted surface testing. Six test panels of the cabinet wall thickness and composition shall be paint finished to withstand testing for adhesion, hardness and salt spray resistance. Three panels shall be prime coated only and three panels shall be prime coated and painted in the identical fashion the cabinet of the equipmcnt item would be accomplished at the factory. The test panels shall be a width of 4-1/2 ± 1/16 inches and a length of 7-1/2 ± 1/16 inches.

4.4.2.11.1 Salt spray. The salt spray test shall be as follows: Three primed panels shall be scribed on one side of each panel with two diagonal scribe lines that extend through the prime coat down to the base metal. For scribing, use a sharp instrument and a suitable straightedge. Hold the scribing tool perpendicular to the surface of the test panel to make two intersecting diagonal scribed marks. Begin each scribed line at a point approximately 1/2 inch from a corner and continue to a point approximately 1/2 inch from the diagonally opposite corner. The scribe shall penetrate the prime coating down to the metal, leaving a uniformly bright line. Expose each metal panel to a salt fog concentration of 5 ± 1 percent, 95 °F with the panel perpendicular to the spray. The panels shall be conditioned in the chamber for two hours before the salt spray fog is applied. The panels shall be exposed for 240 hours. The salt spray fog shall be emitted from an orifice diameter size of between 0.02 and 0.03, inch which atomizes approximately 2.8 liters of salt solution per 10 cubic feet of chamber volume in a 24-hour period. The temperature of the chamber shall be maintained at 95 ± 2 °F and the salt spray shall be maintained within plus or minus 10 °F of the chamber temperature. Remove each test panel from the salt spray cabinet and rinse it off, using a gentle stream of water at a temperature of 100 ± 10 °F. Dry off the test.

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panels and vigorously scrape the scribed side of each panel with a dull knife blade or a piece of sheet metal having a similar dull edge. Hold the scraper with its face perpendicular to the specimen surface and parallel to the scribed lines, moving it back and forth across the scribed lines. Complete the scraping within 15 minutes of the drying of the surface. Masking tape 1-inch wide shall be pressed onto all the scribed marks and pulled away clearly (see 3.6.11.1).

4.4.2.11.2 Hardness. The hardness test shall be as follows: Three primed and painted panels shall be subjected to a hardness test in accordance with ASTM D 3363 using a standard 2H drafting pencil, sharpened to expose the full diameter of the pencil lead. The lead is sanded to a blunt-ended, but sharp-edged shape by using sanding strokes perpendicular to the centerline of the lead. Holding the end of the pencil lead firmly against the enamel test panel with the pencil centerline 45 degrees to the painted surface, apply 15 gouging or scraping strokes to the enameled surface. The strokes shall be a minimum of 1/4 inch long and applied forward, to gouge the lead into the enameled surface (see 3.6.11.2).

4.4.2.11.3 Adhesion. Adhesion between the paint and primer and between the primer and base metal of the panel shall be evaluated by a cross hatch test in accordance with ASTM D 3359 and as follows: The three primed and painted test panels shall be scribed with a sharp knife or similar instrument to give a cross hatch pattern with an overall size of 1.0 square inch. Seventeen parallel cuts shall be made at shout 1/16-inch spacing and extending through the coating to the base metal. Seventeen similar cuts shall be 90 degrees to, and crossing the first 17, giving a cross hatched area of 256 squares. Masking tape 1 inch wide shall be pressed firmly onto the entire hatched area and then immediately jerked cleanly away from the panel. The hatched area shall then be examined (see 3.6.11.3).

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the following (see 6.3):

4.5.1 Inspection lot. A lot shall consist of all machines of each type and class offered for delivery at one time.

4.5.2 Sampling for visual and mechanical inspections. Visual and mechanical examination of the end item shall be for defects as specified in table V and table VI.

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TABLE VI. Sampling for visual and mechanical inspections.

Lot size[1]	Sample size[2]	Accept	Reject[3]
	Major defects		
2 - 5	ALL	0	1
6 - 50	8	0	1
51 - 90	13	1	2
91 - 150	20	1	2
151 - 280	32	2	3
281 - 500	50	3	4
501 - 1200	80	5	6
1201 - 3200	125	7	8
	Minor defects		
3 - 15	3	0	1
16 - 25	5	0	1
26 - 50	8	0	1
51 - 90	13	1	2
91 - 150	20	2	3
151 - 280	32	3	4
281 - 500	50	5	6
501 - 1200	80	8	9
1201 - 3200	125	14	15

[1]All defective items must be replaced with acceptable items prior to lot acceptance.
[2]Inspect sample size until reject criteria is reached.
[3]Rejected lots may be screened and resubmitted for inspection and retest.

4.5.3 Sampling for operational functional, and pressure tests. The test specified in 4.5.3.1, 4.5.3.2, and 4.5.3.3 shall be performed. If any machine fails any test. the entire lot shall be rejected (see table VII).

TABLE VII. Sampling for operational, functional, and pressure tests.

Lot size[1]	Sample size[2]	Accept	Reject[3]
13 - 15	3	0	1
16 - 50	5	0	1
51 - 150	7	0	1
151 - 500	11	0	1
501 - 3200	18	0	1
3201 - 10000	22	0	1

[1]All defective items must be replaced with acceptable items prior to lot acceptance
[2]Inspect sample size until reject criteria is reached.
[3]Rejected lots may be screened and resubmitted for inspection and retest.

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4.5.3.1 Steam pressure. Assemblies of pipes, fittings, coils, cylinders, chests, and chambers using steam shall be tested hydrostatically at 150 lb/in^2 gauge for not less than 30 minutes and the flexible steam condensate and supply hoses shall be tested hydrostatically at 400 lb/in^2 gauge for not less than 30 minutes. No leaks or ruptures shall occur.

4.5.3.2 Functional tests. Functional tests for type I and II presses and type III ironers shall be in accordance with 4.4.2.6.

4.5.3.3 Pneumatic pressure. Pneumatic systems shall be tested at 130 lb/in^2 gauge static air pressure for not less than 30 minutes. No leaks or ruptures shall occur.

4.5.4 Rejection. Failure to pass any of the tests of 4.5.3.1, 4.5.3.2, and 4.5.3.3 shall be cause for rejection of the unit.

4.6 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

5.1 General.

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.6.)

a. *Navy shipboard stowage fire-retardant requirements.*

(1) Treated lumber and plywood. Unless otherwise specified (see 6.2.1), all lumber and plywood including laminated veneer material used in shipping containers and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B	-	Type II - weather resistant Category 1 - general use
Level C	-	Type I - non-weather resistant Catetory 1 - general use.

(2) *Fiberboard.* Unless otherwise specified (see 6.2.1), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, cleated fiberboard boxes including interior packaging forms shall meet the flame spread index and the specific optic density requirements of PPP-F-320 and amendments thereto.

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5.2 Preservation, packing, and marking. Laundry presses and finishing equipment shall be preserved level A or commercial, packed level A, B, or commercial, and marked in accordance with MIL-L3153 as specified (see 6.2.1). Marking shall include bar codes and applicable acquisition options herein as specified (see 6.2.1). In addition, equipment shall be shipped fully assembled, unless disassembly is necessary to protect certain components during transit, or disassembly results in more than a 25 percent savings in transportation costs to the Government.

5.3 Technical manuals. Technical manuals which accompany shipments shall be packaged in a transparent waterproof plastic bag, minimum four mil thick. Closure shall be by heat sealing. Technical manuals shall not be placed with any flexible sealed barrier enclosing components. The copy(s) of the manual shall be placed in the shipping container housing the main unit. Packing lists shall indicate which container contains the technical manual(s) and shall state the approximate location therein. The manual shall be readily accessible when the container is opened. Technical manuals, when shipped in bulk quantities, shall not be individually wrapped, but shall be packed in accordance with the requirements of the applicable technical manual specifications.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Commercial laundry pressing and finishing equipments acquired under this specification are intended for shipboard use. Therefore, commercial machines offered to the Government may require alteration to meet specification requirements, for example, mounting provisions; electrical power; and materials.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification
b. Type and class of item required (see 1.2)
c. When first article inspection is required (see 3.1)
d. Dismantling requirements (see 3.6.2.3.15)
e. Whether technical manuals are required (see 6.3)
f. If test conditions are other than specified (see 4.3)
g. Fire-retardant material requirements (see 5.1)
h. Preservation, packing, and marking requirements (see 5.2).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR supplement 27.475-1 exempts the requirement for a DD Form 1423.

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Referenced paragraph	DID number	DID title	Suggested tailoring
3.5	DI-E-7031	Drawings, engineering and associated lists	-
4.5	DI-T-5329	Inspection and test reports	-
4.4	DI-T-4902	First article inspection report.	-

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Technical manuals The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals arc required, military specifications and standards that have been cleared and listed in the DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 First article. When the first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items(s) should be a preprpduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation

6.6 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.7. When ordering spare parts or repair parts for the equipment covered by this specification. the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacturer of the equipment Packaging for such parts should also be specified.

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6.8 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

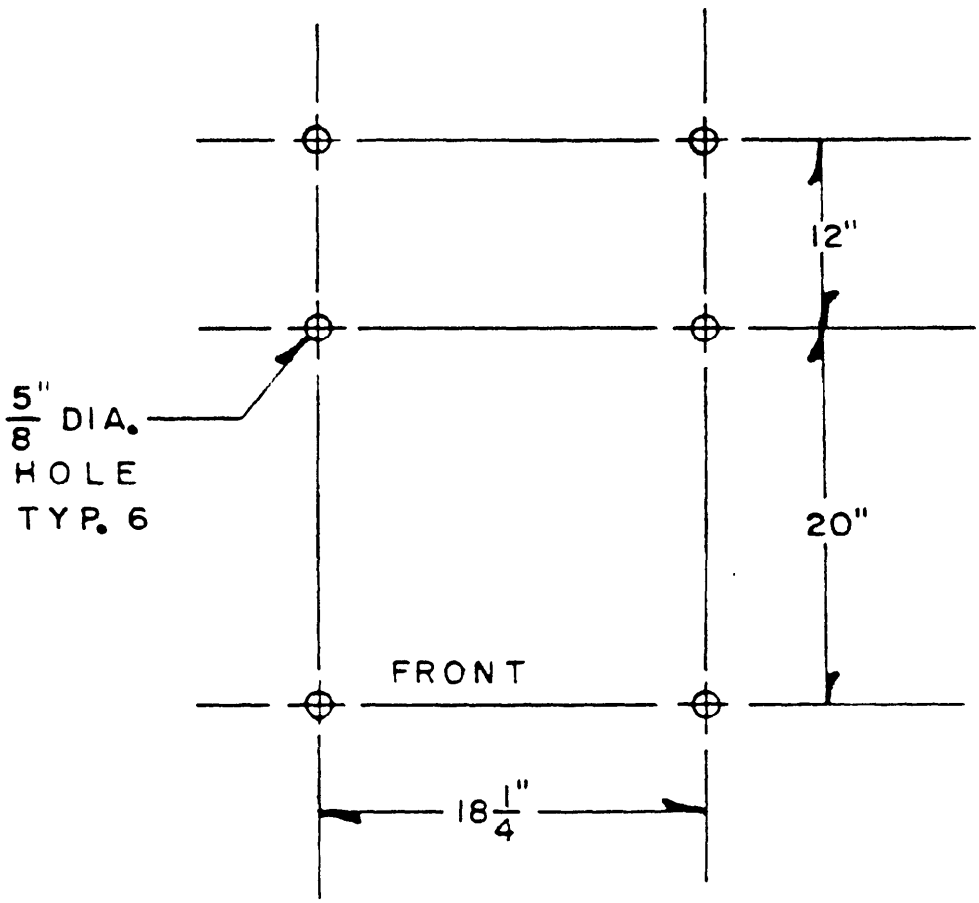
6.9 Subject term (key word) listing.

Electrical power, materials
Ironer
Press control

6.10 Changes from preview issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 3510-N287)

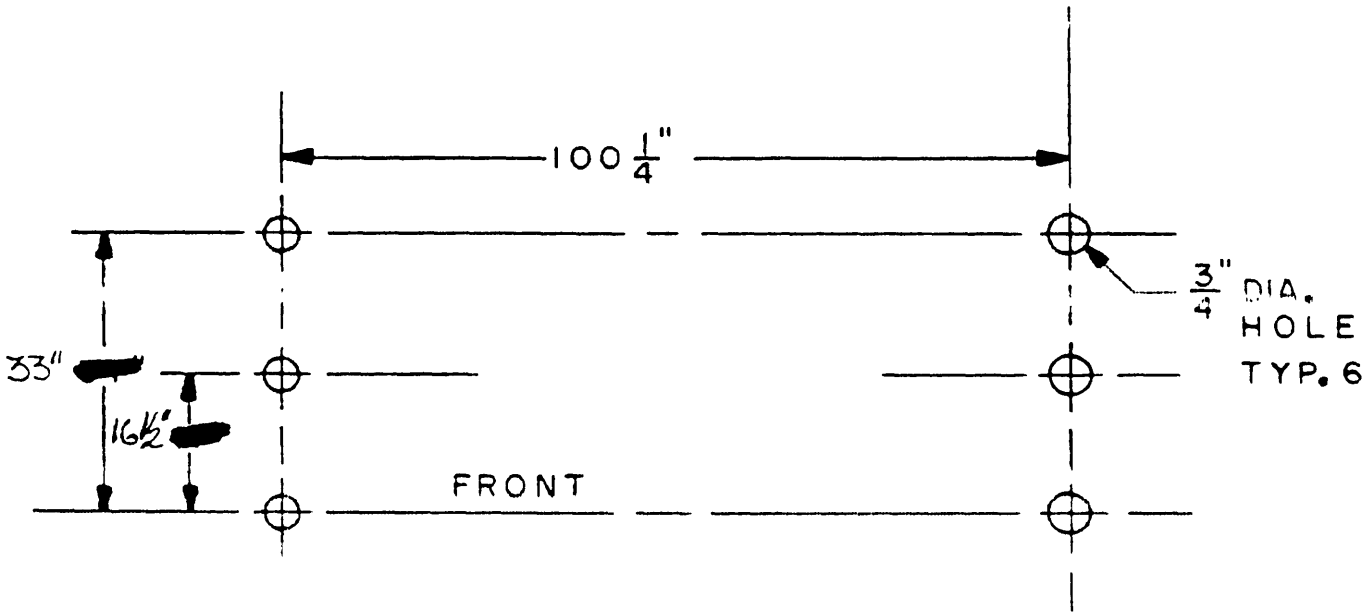
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SH 13231756

FIGURE 1. Laundry presses.

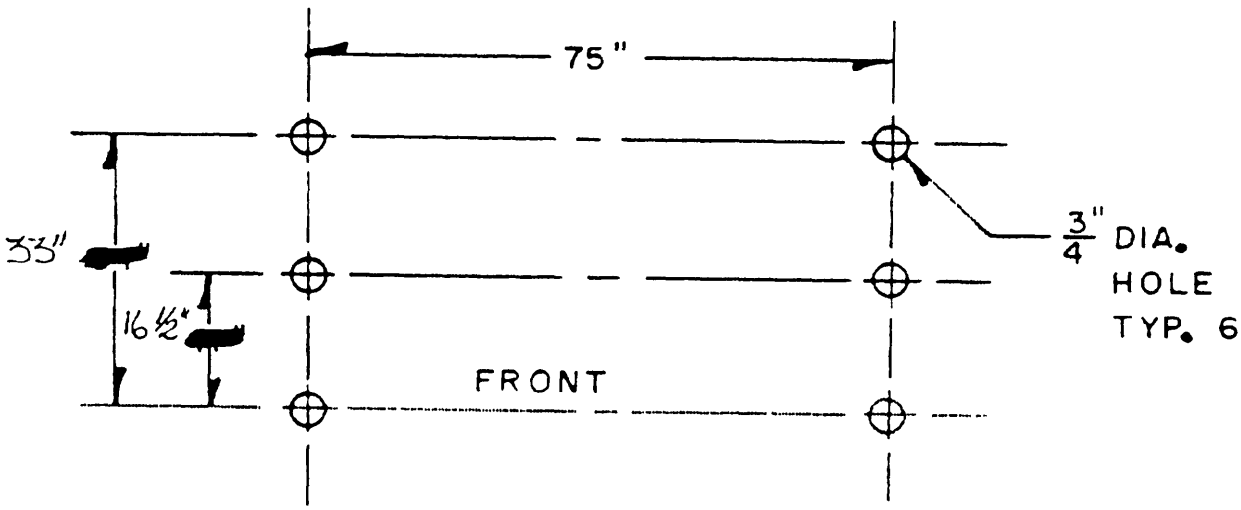
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SH 13231757

FIGURE 2. Ironer – 85 inch.

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FIGURE 3. Ironer – 60 inch.

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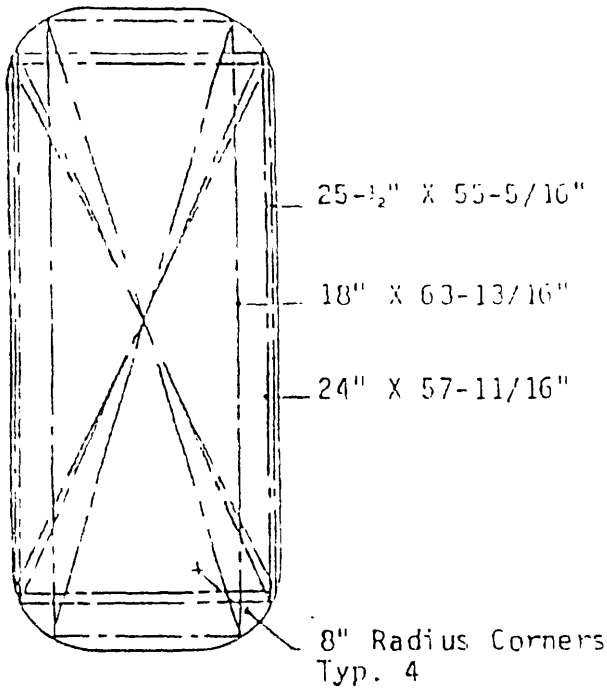


FIGURE 4. 26” X 66 watertight door.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE *(YYMMDD)*

DOCUMENT TITLE

NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

REASON FOR RECOMMENDATION

SUBMITTER

NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

PREPARING ACTIVITY

NA[illegible] Technical Point of Cotact (TPOC): Mr. I. Gublis (SEA 56Z44) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS	b TELEPHONE *(Include Area Code)* (1) Commercial TPOC: 703-602-6065	(2) AUTOVON 8-332-6065
ADDRESS *(Include Zip Code)* Commander, Naval Sea Systems Command Department of the Navy SEA 55Z3) Washington, DC 20362-5101	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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